Machine Learning

Support Vector Machines: Support Vector Classifier

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Support vector classifier

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- The `support vector classifier` is a `soft margin classifier` that does not aim to separate the classes perfectly.

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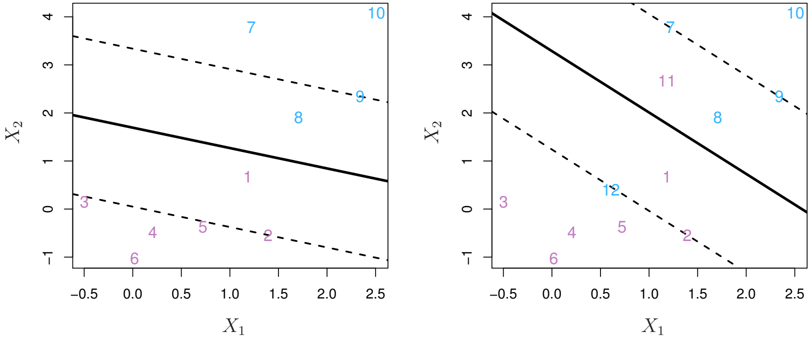
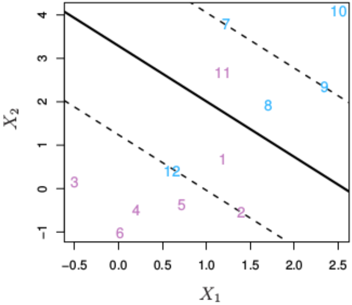
- The maximal margin classifier tries to separate the classes perfectly.
- The support vector classifier is a soft margin classifier that does not aim to separate the classes perfectly.
- It allows a “budget” to have observations on the wrong side of the margins, or even on the wrong side of the hyperplane, in order to
  - have greater robustness to individual observations;
  - do better classification on most of the observations;
  - allow to find a solution in the non-separable case.



Support vector classifier



$$\underset{\beta_0,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n}{\text{maximize}} \quad M$$

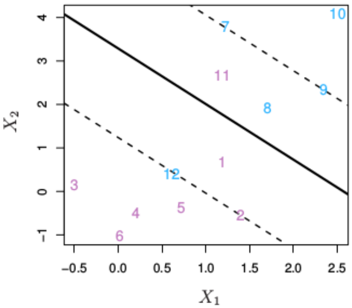
$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \sum_{i=1}^{n} \epsilon_i \leq b \quad \text{for all } i = 1, \ldots, n.$$

- The "budget" ϵ_i tells us for each observation x_i by `how much it violates the margin`: it is on the correct side of the margin if $\epsilon_i = 0$; on the wrong side of the margin if $\epsilon_i > 0$; on the wrong side of the hyperplane if $\epsilon_i > 1$.
- The constant $b > 0$ is a `tuning/regularization` parameter, the total budget for the amount that the margin can be violated by the n observations.

Support vector classifier



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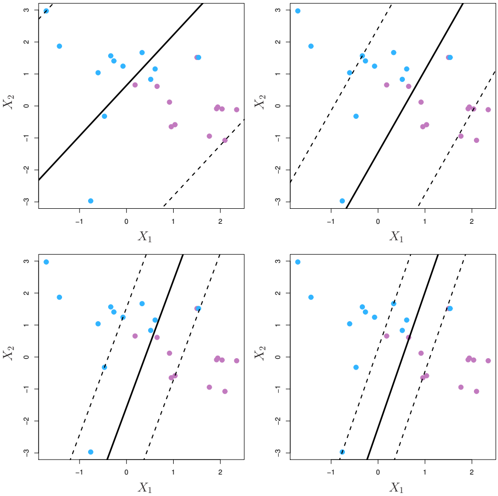
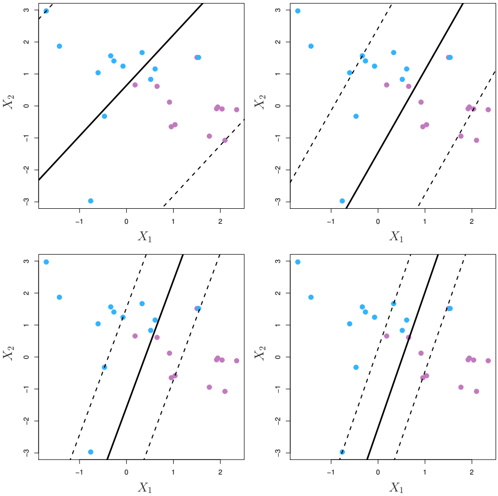
- The constant $b > 0$ is a `tuning/regularization` parameter, the total budget for the amount that the margin can be violated by the n observations.
- Here, the `support vectors` are those who are on the wrong side of the margin for their class ($\epsilon_i > 0$) or lie directly on the margin, and only they affect the classifier!
- Thus, b controls the number of support vectors, and therefore the `bias-variance trade-off` of the support vector classifier.

Support vector classifier: the tuning parameter b

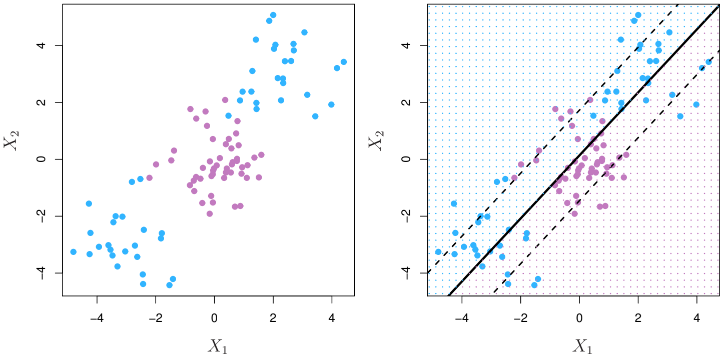


Support vector classifier: the tuning parameter b



- Large b gives `wide margins`, and `high bias/small variance` for the SV classifier
- Small b gives `tight margins`, and `small bias/high variance` for the SV classifier.
- The optimal b can be found by `cross-validation`.

Support vector classifier: limitations



- A `linear decision boundary` can fail if the true boundary between the classes is not linear in the feature space of $(X_1, \ldots, X_p)$! Any linear classifier will perform poorly here.

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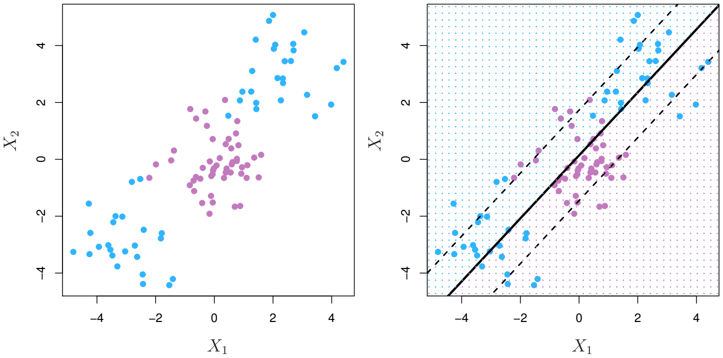

- A linear decision boundary can fail if the true boundary between the classes is not linear in the feature space of $(X_1, \ldots, X_p)$! Any linear classifier will perform poorly here.
- One might want to enlarge the feature space, say with order-two polynomials $X_1, X_1^2, \ldots, X_p, X_p^2$, and then use a SV classifier with the $2p$ features.
- There are many ways to enlarge the space, and the huge number of features will make the computations infeasible.